

Swopnab Bikram Karki

+1 986-600-8163 | swopnabbikram@gmail.com | linkedin.com/in/swopnabbkarki | github.com/Swopnab

EDUCATION

University of Texas at Arlington
B.S. in Computer Science

Arlington, TX
Expected Graduation: Fall 2027

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

Web & Backend: React, Node.js, Django, Django REST Framework, Express.js, Flask, Hono, REST APIs, HTML5, CSS3, HTML5 Canvas, Vite, API Integration

AI / Machine Learning: Large Language Models (LLMs), PyTorch, TensorFlow, Scikit-learn, OpenAI API, LangChain, RAG, Vector Databases, Google MediaPipe, Ollama, Llama 3.2, Generative AI, AI Assistants, Retrieval-Augmented Generation (RAG)

Cloud & Databases: AWS (S3, IAM), Cloudflare Workers, Cloudflare D1, SQLite, GitHub Pages

Security: JWT Authentication, RBAC, Refresh Token Rotation, PBKDF2-SHA256, Web Crypto API, Session Management, Per-User Data Isolation

Tools: Git, GitHub, GitHub Actions, Wrangler, Docker, Kubernetes, VS Code, Postman, Linux, Figma, CI/CD Tools

PROJECTS

SwopMobile – JWT Authentication, RBAC & AI Workspace

2025–2026

Full-Stack Authentication and Multi-App Platform

JavaScript, Cloudflare Workers, Hono, D1, Flask, Ollama

Built a serverless authentication platform supporting 3 RBAC roles with JWT access tokens, refresh-token rotation, protected routes, and per-user data isolation using Cloudflare Workers and D1.

Improved account security with PBKDF2-SHA256 password hashing, 15-minute access tokens, tab-scoped sessions, and server-side permission enforcement.

Integrated Zabum AI using Flask, Ollama, and Llama 3.2 with authenticated chat history, persistent memory, and permission-based access.

CyberSteer – Hand-Controlled Racing Game

2026

Computer Vision / Browser Game

JavaScript, HTML5 Canvas, MediaPipe Hands, CSS3

Built a real-time browser racing game by mapping 21 MediaPipe hand landmarks into left, center, and right steering controls.

Implemented procedural traffic, AABB collision detection, smooth steering, and speed progression from 50 to 100 mph with persistent high scores.

AAURA-V1 – Advanced AI-Utilized Responsive Assistant

2024

Full-Stack AI Application

Python, Django, React, TypeScript, AWS, OpenAI API

Built a full-stack AI assistant with Django REST Framework, React, JWT authentication, and OpenAI API integration.

Enabled secure cloud file storage and access control using AWS S3 and IAM.

Personal Job Tracker

2025

Job Application Tracking Web App

JavaScript, HTML, CSS, GitHub Pages

Built a responsive web app for tracking companies, job roles, and application statuses using vanilla JavaScript.

Deployed the application through GitHub Pages with reusable front-end components and persistent browser-based data.

CERTIFICATIONS

Microsoft Azure Essentials Professional Certificate by Microsoft and LinkedIn

Career Essentials in Software Development by Microsoft and LinkedIn

Career Essentials in GitHub Professional Certificate

Excel Essential Training (Microsoft 365)